

HGSOC Appendix (All Plots & Figures)

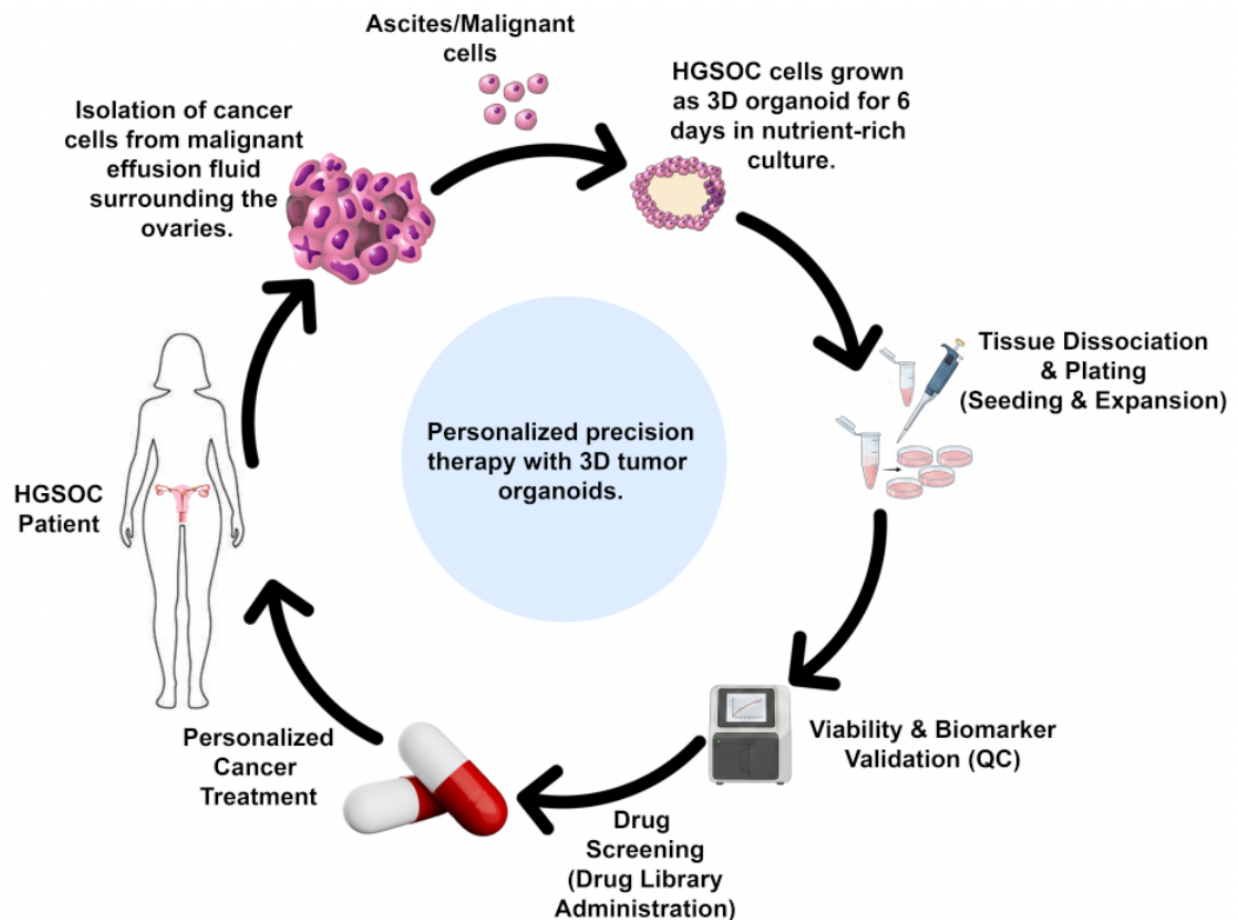


Figure 1. Schematic workflow of a patient-derived 3D organoid pipeline for functional precision medicine in HGSOC. Malignant cells are isolated from the ascites or fluid surrounding the ovaries of a High-Grade Serous Ovarian Carcinoma (HGSOC) patient. The cells are cultured as 3D tumor organoids for 6 days in a nutrient-rich medium to establish structural and genomic architecture. Following initial culture, organoids undergo tissue dissociation and are replated into multi-well plates for seeding and expansion. Automated phenotypic quality control (QC) validates cell viability and biomarker expression prior to high-throughput drug library screening. The resulting therapeutic profile translates directly into personalized cancer treatments for the patient.

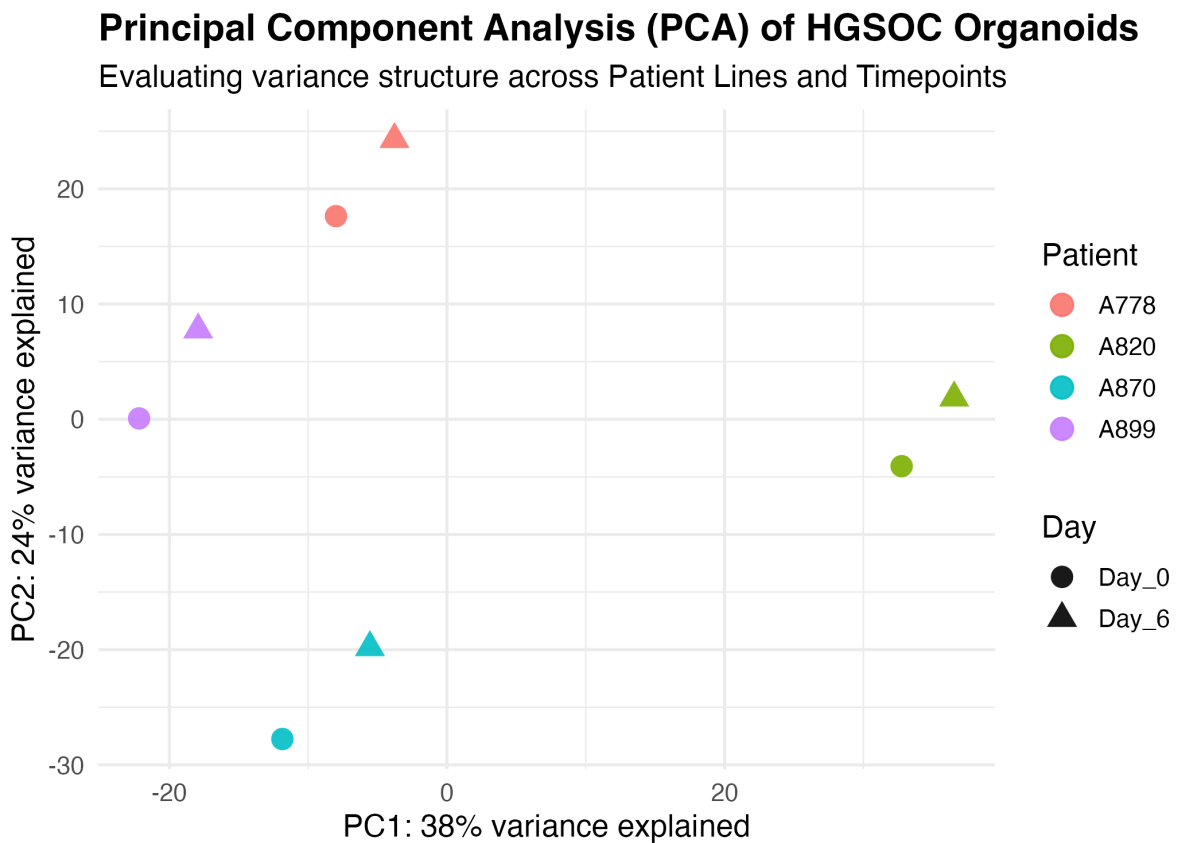


Figure 5. Principal Component Analysis (PCA) of HGSOC Organoids. PCA of variance-stabilized RNA-seq data showing global transcriptomic variation. Samples are coloured by patient and shaped by timepoint (Day 0 vs. Day 6). **Strong clustering by patient indicates that patient-specific differences are the primary source of variation.**

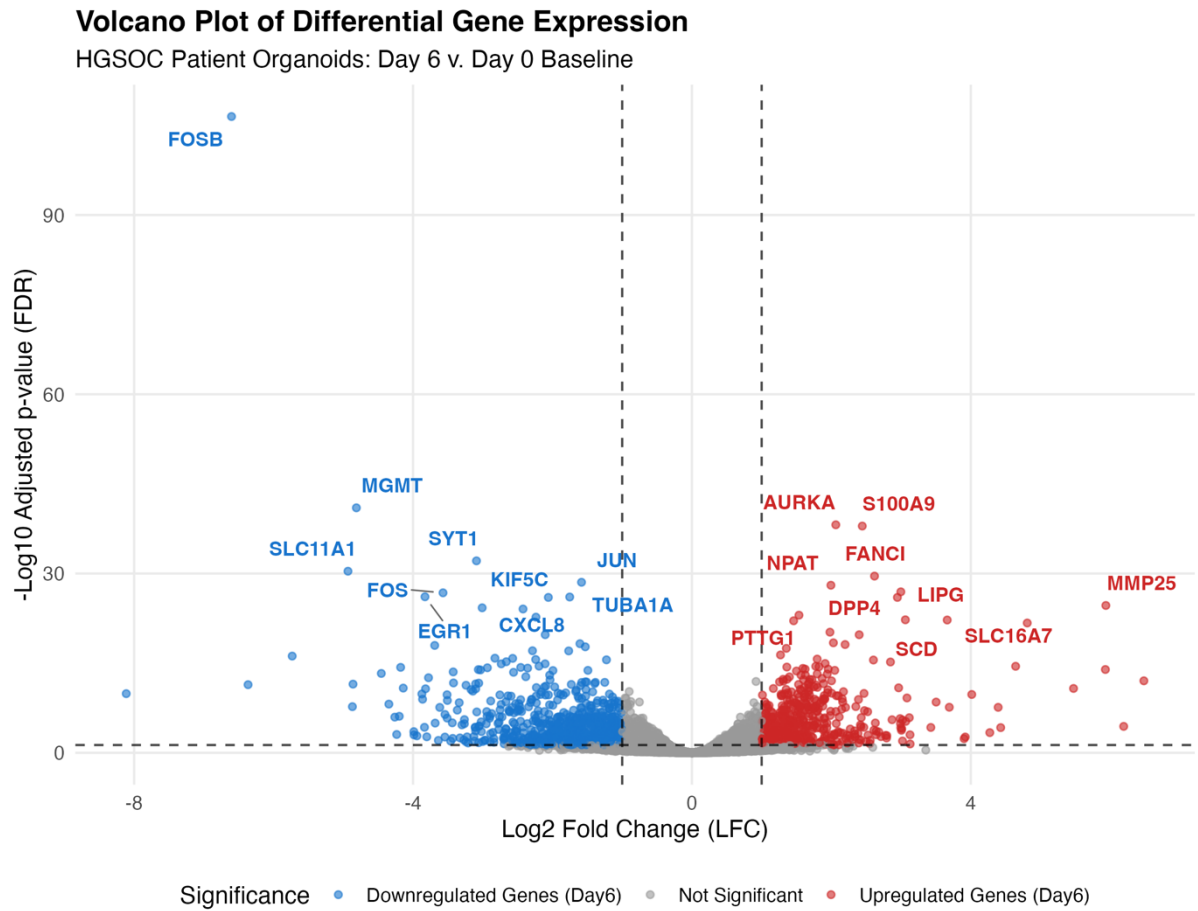


Figure 6. Volcano Plot of Differential Gene Expression (Day 6 vs. Day 0 baseline) across all HGSOC patient-derived organoids. Each point represents a gene, illustrating the overall transcriptomic changes associated with organoid culture over time.

Top 100 Differentially Expressed Genes - Day 6 v. Day 0 Baseline

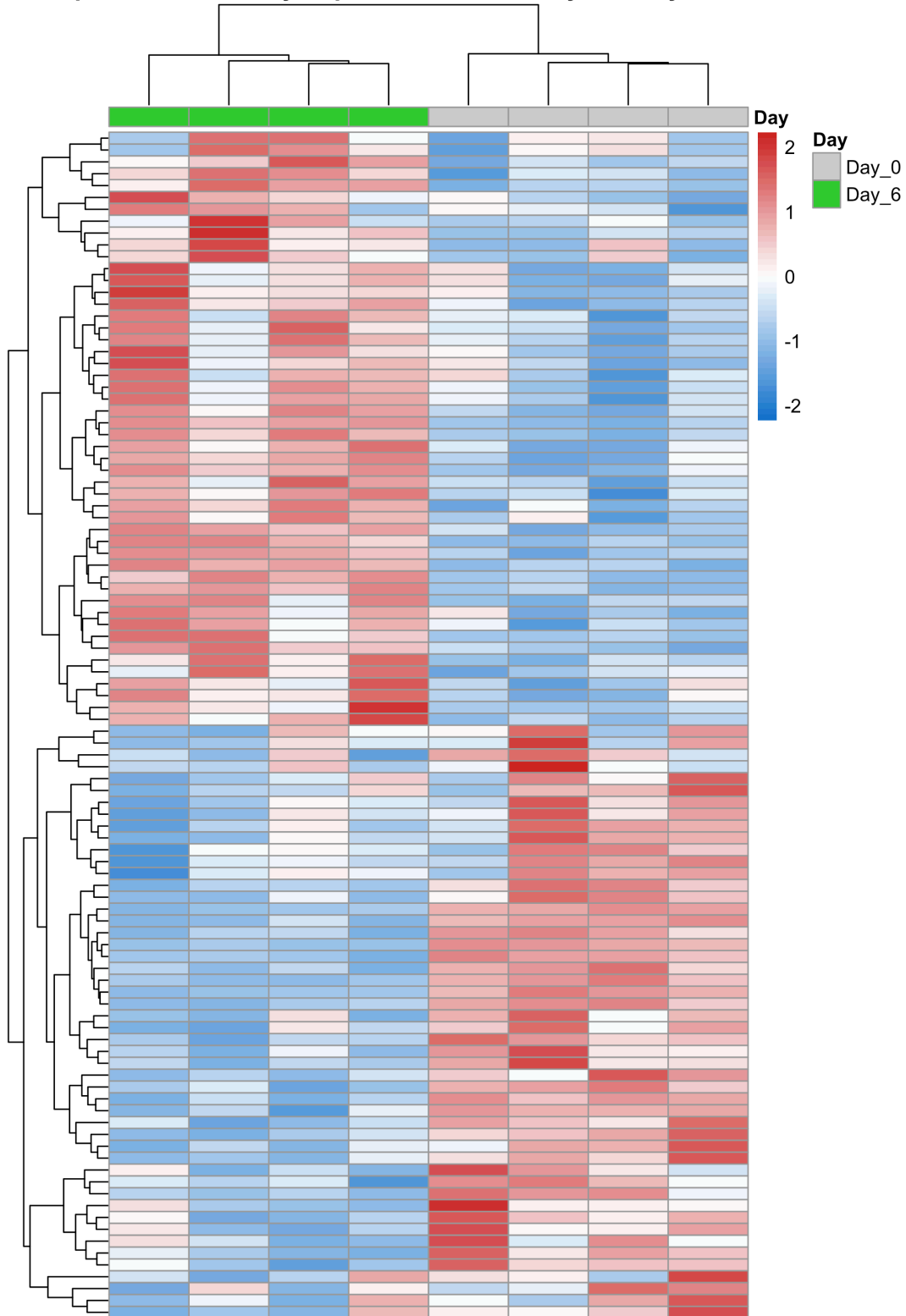


Figure 7. Heatmap of the Top 100 Differentially Expressed Genes (Day 6 vs. Day 0). Samples cluster perfectly by timepoint (top dendrogram). The genes split into two clear blocks: ~60% upregulated and ~40% downregulated at Day 6 (left dendrogram). Colours indicate relative expression variations (Z-scores). **Deep Red (+2)** = Upregulated, **White (0)** = Average Expression, **Deep Blue (-2)**: Downregulated.

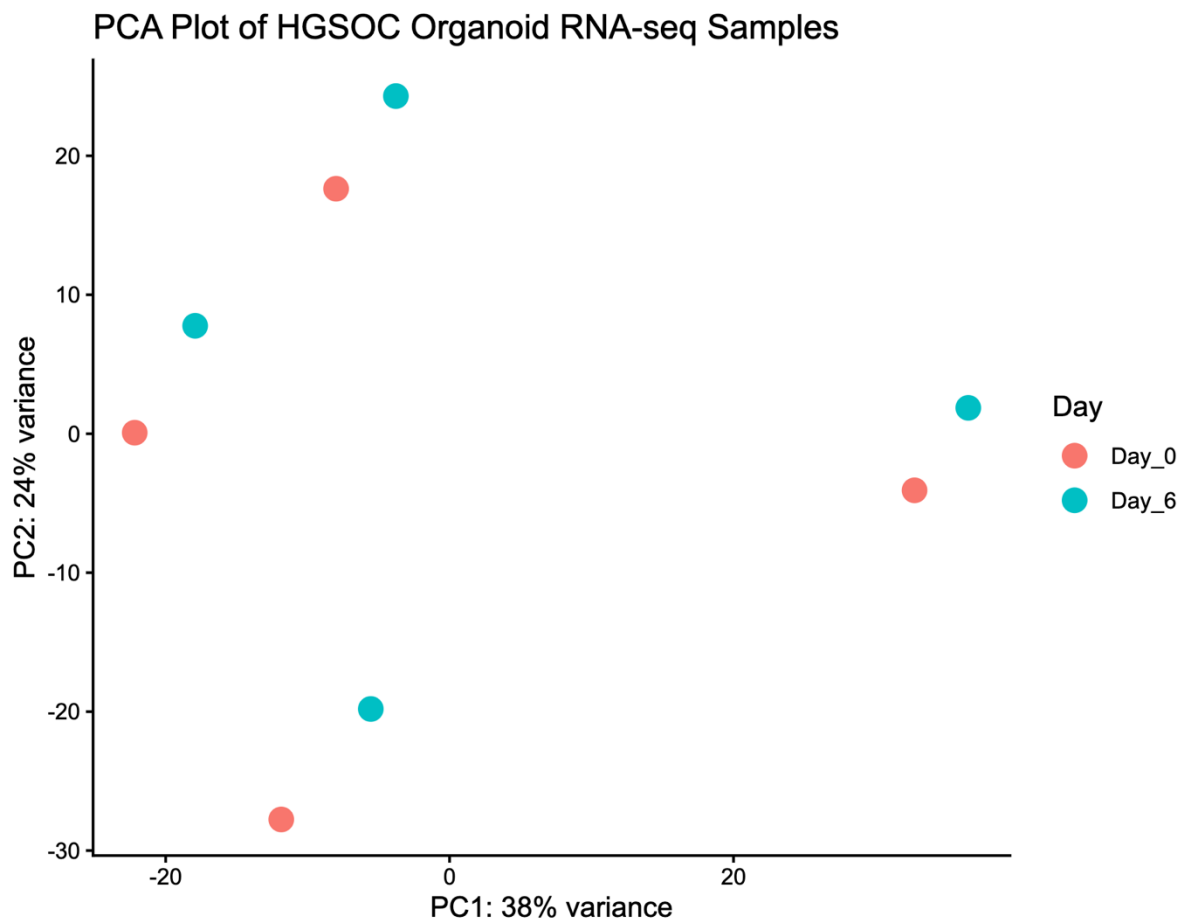


Figure 8. QC Principal Component Analysis (PCA) of HGSOC organoid transcriptomes. Variance-stabilized RNA-Seq read counts across the four patient-derived lines are projected onto the first two principal components, which together account for 62% of the total transcriptional variance (PC1: 38%; PC2: 24%). Individual samples are colour-coded by culture timepoint: immediate post-isolation baseline (Day 0, salmon) and after a 6-day *in vitro* culture period (Day 6, cyan). In alignment with the hierarchical clustering results, samples group tightly into four distinct coordinate spaces corresponding to individual patient genetic backgrounds. The proximity of Day 0 and Day 6 pairs for each unique patient line relative to the vast spatial separation between patient groups demonstrates that inter-patient biological heterogeneity remains the primary driver of transcriptomic variation, exceeding the gene expression shifts induced by the 6-day culture period.

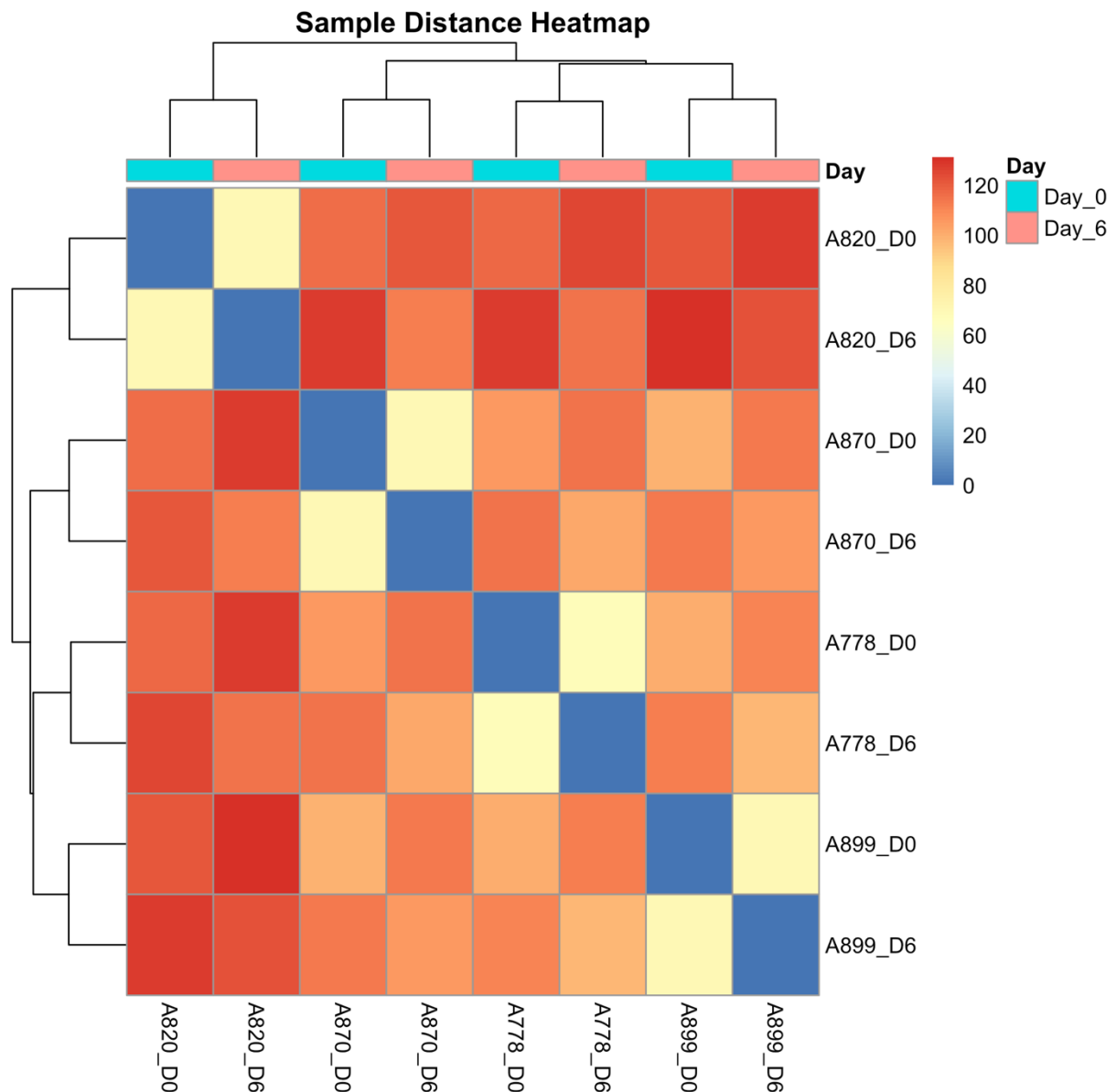


Figure 9. Sample distance heatmap of HGSOc organoid transcriptomes across experimental conditions. Hierarchical clustering analysis based on Euclidean distances computed from variance-stabilized RNA-Seq read counts across four patient-derived lines (A778, A820, A870, and A899). The colour gradient indicates relative distance, where darker blue signifies high transcriptional similarity (lower distance) and red signifies lower similarity (higher distance). The dendrogram demonstrates that samples cluster primarily by patient genetic background rather than by treatment timepoint (Day 0, cyan; Day 6, salmon). For each independent patient line, the Day 0 and Day 6 cultures cluster directly together as immediate sister pairs, highlighting that inter-patient biological variability outweighs the transcriptomic shifts induced by the 6-day *in vitro* culture microenvironment.

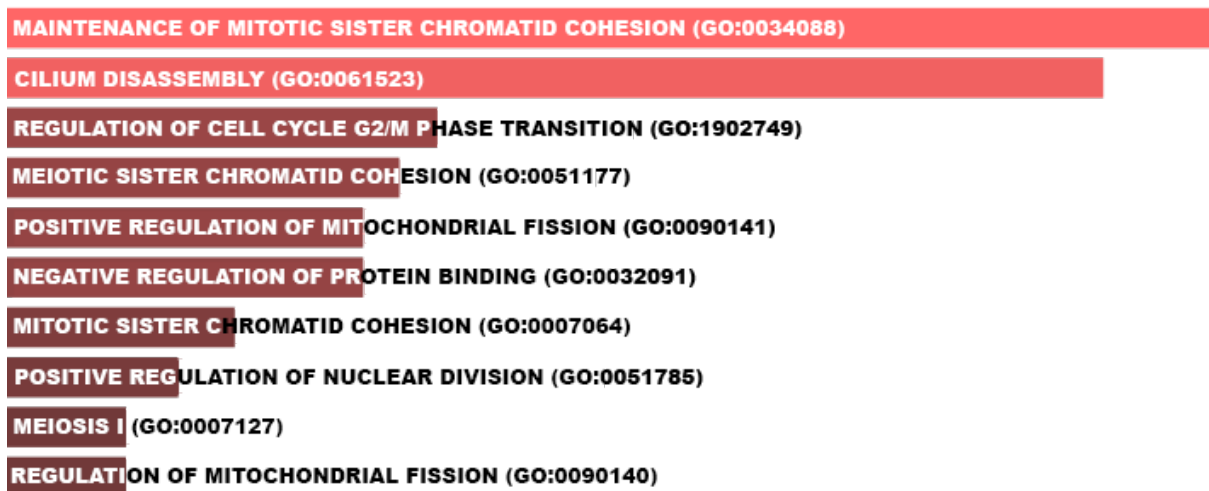


Figure 10. Gene Ontology Enrichment Analysis of Upregulated Genes (Day 6). Top upregulated biological processes, showing exit from G0 (cilium disassembly) and progression into active mitosis (chromosome cohesion, G2/M transition). Essentially, The cells exited quiescence and entered a phase of rapid proliferation.

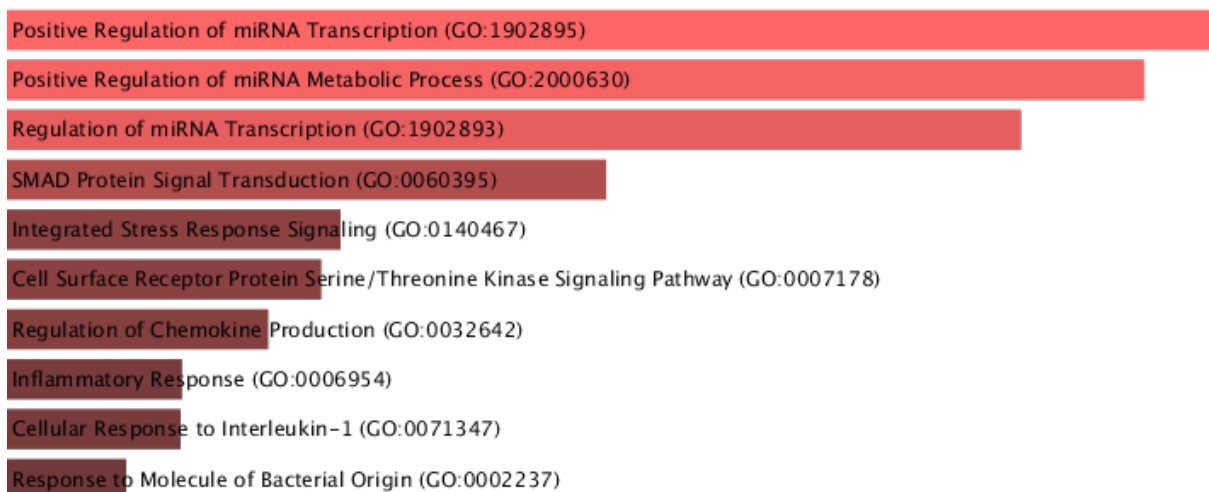


Figure 11. Gene Ontology Enrichment Analysis of Downregulated Genes (Day 6). Enriched GO terms confirm the loss of host microenvironmental stress and immune signaling, characterized by significant reductions in integrated stress response signaling, inflammatory response, and chemokine production. This confirms the exit of the cells from an acute inflammatory and stressed state (characteristic of patient malignant effusion fluid).